

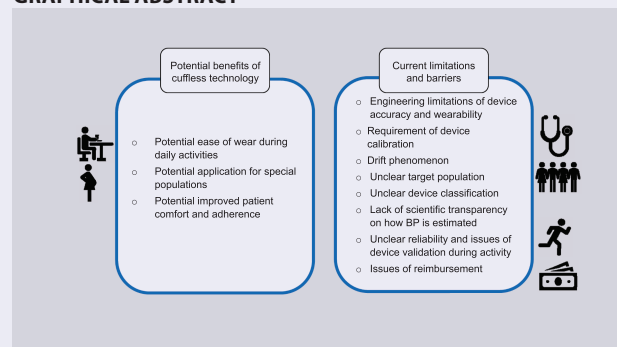
Cuffless Blood Pressure Devices

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Hypertension is associated with more end-organ damage, cardiovascular events, and disability-adjusted life years lost in the United States compared with all other modifiable risk factors. Several guidelines and scientific statements now endorse the use of out-of-office blood pressure (BP) monitoring with ambulatory BP monitoring or home BP monitoring to confirm or exclude hypertension status based on office BP measurement. Current ambulatory or home BP monitoring devices have been reliant on the placement of a BP cuff, typically on the upper arm, to measure BP. There are numerous limitations to this approach. Cuff-based BP may not be well-tolerated for repeated measurements as is utilized with ambulatory BP monitoring. Furthermore, improper technique, including incorrect cuff placement or use of the wrong cuff size, may lead to erroneous readings, affecting diagnosis and management of hypertension. Compared with devices that utilize a cuff, cuffless BP devices may overcome challenges related to technique, tolerability, and overall utility in the outpatient setting. However, cuffless devices have several potential limitations that limit its routine use for the diagnosis and management of hypertension. The review discusses the different approaches for determining BP using various cuffless devices including engineering aspects of cuffless device technologies, validation protocols to test accuracy of cuffless devices, potential barriers to widespread implementation, and future areas of research.

This review is intended for the clinicians who utilize out-of-office BP monitoring for the diagnosis and management of hypertension.

GRAPHICAL ABSTRACT



Keywords: ambulatory blood pressure monitoring, blood pressure, blood pressure determination, blood pressure monitors, hypertension

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The diagnosis and management of hypertension (HTN) has been traditionally based on in-office blood pressure (BP) measurement¹; however, recent guidelines have emphasized out-of-office BP monitoring and have the potential to impact BP management in primary care.² Furthermore, practical challenges to physical assessment wrought by the COVID-19 pandemic have increased awareness of and interest in strategies for out-of-office BP monitoring.³ Out-of-office BP monitoring includes ambulatory BP monitoring (ABPM) and home BP monitoring (HBPM).⁴ The vast majority of current ABPM and HBPM devices use the oscillometric method, which involves a BP cuff typically placed on the upper arm. While cuff-based BP measurement devices are sufficient for single measurements, they may be not well-tolerated for frequent, repeated measurements such as those taken over a 24-hour period as with ABPM.¹ Furthermore, the BP cuffs may be challenging for patients to use correctly, which can

result in improper technique. In addition to incorrect cuff placement, the use of the wrong cuff size can lead to erroneous BP readings, which can greatly impact the diagnosis and management of HTN.⁵

“Cuffless” BP devices that do not rely on a BP cuff have emerged as a promising and novel method of BP estimation. A cuffless BP technology could eliminate issues such as patient intolerance and improper cuff placement or size. With increased convenience and tolerability, the use of cuffless BP devices may also improve adherence to out-of-office BP monitoring. In recent years, there has been a substantial increase in the number of patents for cuffless BP devices and devices sold on the market.⁶ However, there are several major concerns about the accuracy and utility of these devices that have not been adequately addressed. Understanding how these devices work and their limitations are essential before implementing cuffless BP technologies in routine clinical practice.

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CURRENT TECHNIQUES

The description of devices discussed herein is not exhaustive and was selected to illustrate the different available technologies. None of the approaches currently used by cuffless BP devices directly measure BP (Table 1). Instead, BP is estimated by directly measuring other physiologic variables and using those measurements to estimate BP based on predetermined algorithms. There are 2 main variables that are used to measure BP: the time required for pressure waves to travel between 2 points and the amplitude of the arterial waveform. After measuring one or more of these variables, cuffless BP devices use several approaches to estimate BP. The most common approaches include pulse transit time (PTT), pulse arrival time (PAT), tonometry, and pulse wave analysis (PWA). Many of these approaches coexist within the same device.

Pulse transit time

Pulse transit time is defined as the time it takes for an arterial pressure wave to travel between 2 sites (i.e., from the heart to a peripheral site) and is inversely related to BP.⁷⁻⁹ PTT is used to calculate BP by first measuring the time required for an arterial waveform to propagate between a proximal and distal site of measurement, which can then be used to determine pulse wave velocity (PWV), an indicator of arterial stiffness that has been extensively studied.¹⁰⁻¹² PWV is then related to BP by applying the Moens-Korteweg¹³ and Hughes¹⁴ equations, which relate PWV to vascular stiffness and vascular stiffness to BP, respectively. By combining these 2 equations, one can create a simple logarithmic model relating BP to PWV. Because the time required for the wave to travel depends on the viscoelastic properties of the arterial wall for each person, individualized calibration of the measured PTT to BP for a person, determined using a known reference standard, which is commonly an oscillometric device with a BP cuff, is required.¹⁵ As PWV is derived from PTT using a known distance and is not a measure that is independent from PTT, PWV-based approaches will not be distinguished from PTT-based approaches in this review.

The approach used to measure both the proximal and distal arterial waveform is a distinguishing feature of different PTT-based devices. The most common approach is to use 2 photoplethysmography (PPG) sensors, the same technology used in pulse oximetry, that uses a light-emitting diode (LED) as a light source coupled with a photodetector to measure the amount of light that is absorbed by blood vessels. Light absorption depends on blood volume in the light's path, with an increase in absorption indicating an increase in blood volume and the arrival of the pulse wave. Another sensor that can be used in place of a distal PPG signal is impedance plethysmography. Impedance is related to cross-sectional area of artery; an increased impedance reflects an increased cross-sectional area suggesting the arrival of blood volume and a pulse wave.⁸ Another distinguishing feature is the measurement sites. The 2 locations can be along the same arterial path (e.g., the upper arm and the fingertip), or they can be proximal and distal relative to

one another but not along a contiguous artery (e.g., the forehead and the fingertip).

An example of a PTT device is one by BioBeat (BioBeat, Tel Aviv, Israel), which uses a proprietary arrangement of PPG sensors on the back of a watch face worn on the wrist. In Bland-Altman agreement analyses, mean difference between the BioBeat device and the reference standard among healthy subjects was -0.08 (95% confidence interval [CI] $-7.06, 6.90$) mm Hg for systolic BP (SBP) and 0.002 (95% CI $-6.88, 6.87$) mm Hg for diastolic BP (DBP).¹⁶ There was an absolute greater degree of bias detected in participants with HTN: 0.49 (95% CI $-7.13, 6.14$) mm Hg for SBP and -0.63 (95% CI $-7.11, 5.86$) mm Hg for DBP.¹⁶

This watch, as well as the company's wearable chest patch that utilizes the same technology, has also been compared with a gold standard of an arterial line.¹⁷ However, as this was done in an inpatient, postoperative population rather than an ambulatory outpatient setting, the application to the routine clinical setting remains unclear.

Pulse arrival time

PAT is the sum of the pre-ejection period (PEP) and PTT. The PEP is the time elapsed between the electrical depolarization of the left ventricle and the beginning of ventricular ejection, and it represents the period when the left ventricle is contracting against a closed aortic valve. The R wave on an electrocardiogram (ECG) signal is used to determine the electrical depolarization of the left ventricle.¹⁸ Since the R wave on the ECG signal is used as the proximal site, PAT only requires one peripheral signal, such as PPG, compared with 2 sensors present in a standard PTT device. The mathematical method to estimate BP from PTT is similarly applied to a measured PAT. However, in contrast to using PTT, when PAT is used to estimate BP, the device's algorithm makes assumptions about how much PEP contributes to PAT. Because the duration of PEP depends on an individual's specific physiologic conditions (i.e., the presence of sympathetic stimuli, abnormal cardiac conduction, or abnormal myocardial contractility), the contribution of the PEP to PAT varies by individual. As a result, PAT may have increased error compared with using PTT alone.¹⁸

In part to overcome potential error introduced by PEP, several devices have attempted to find an alternative for the proximal ECG component of the PAT approach. An emerging option is to utilize technologies embedded in smartphones. The Instant BP app (IBP; AuraLife, Aura Labs, CA) estimates BP using a technique in which the top edge of the smartphone is placed on the left side of the chest while the individual places his or her right index finger over the smartphone's camera.¹⁹ The application uses proprietary vascular indices and the phone's microphone to capture heart sounds as a replacement for detecting the R wave on an ECG, while the smartphone camera functions as a distal PPG sensor. However, this technique was found to be inaccurate and was removed from the market, indicating that, while novel, this approach is not ready for clinical use.¹⁹ Chandrasekaran *et al.* proposed a similar method, replacing the R-wave on the ECG with an external microphone and using the phone's camera as a distal PPG.²⁰

Table 1. Methods of estimating BP using cuffless BP technologies

	PTT		PAT	Tonometry		PWA
Variable measured	Time between 2 points of pressure wave propagation (e.g., the upper arm and the finger); PWV can be determined from this value		Time between R wave of ECG to a distal point of pressure wave propagation	Pressure waveform		Raw signal or salient features from arterial pulse waveform
Common sensors	PPG		PPG and ECG	Pressure sensor		PPG and radar
Assumptions	Blood density Arterial geometry Vascular stiffness		Blood density Arterial geometry Vascular stiffness	MAP is determined and SBP and DBP are derived according to the device's algorithm		Arterial geometry, vascular stiffness, and location of wave reflection sites (hemodynamics theory-based methods) Waveform morphology is predictive of BP; modeling assumptions depend on algorithm choice (AI-based methods)
Advantages	Smartphone capability Flexibility in sensor location		Smartphone capability ECG technology widely used	Can be single site		Can be single site Considers multiple physiologic variables, including the effect of branchpoints in the vascular system and the contractile forces of the heart Can combine with other methods to improve accuracy
Disadvantages	Requires at least 2 sensors		Requires at least 2 sensors PEP is a source of inaccuracy May be difficult to integrate continuous ECG in unobtrusive form	Requires firm appplanation to artery		May be more sensitive to motion artifacts and other noise

Abbreviations: DBP = diastolic blood pressure; MAP = mean arterial pressure; PAT = pulse arrival time; PEP = pre-ejection period; PPG = photoplethysmography; PTT = pulse transit time; PWA = pulse wave analysis; SBP = systolic blood pressure.

Tonometry

Applanation tonometry involves placing direct pressure on the skin overlying a peripheral artery to obtain an aortic waveform tracing.²¹ With the tonometry sensor exerting a downward force on the targeted artery, the force of the wall tension of the artery is directed perpendicularly, directing the force of the arterial BP toward the sensor. Using the detected waveform tracing, the sensor then determines the pulse pressure and provides an estimate of BP.²² An example of this technology is the BPro device (HealthSTATS Technologies, London, England).²³ This device uses radial artery applanation in a watch strap to maintain constant and substantial pressure on the radial artery, which can be uncomfortable. Validation studies were performed with a standard ABPM oscillometric cuff-based device as the reference standard. In a study of 50 participants without HTN (SBP/DBP <140/90 mm Hg), the Bland–Altman plots of the BP values provided by the BPro versus the reference standard showed a bias of -4.5 mm Hg (95% CI -27.3 mm Hg, 18.3 mm Hg) for SBP values and -0.8 mm Hg (95% CI -17.8 mm Hg, 16.2 mm Hg) for DBP during the awake period.²³ For the sleep period, the authors reported a bias of 2.1 mmHg (95% CI -27.1 mm Hg, 31.1 mm Hg) for SBP and -7.4 mmHg (95% CI -11.0 mm Hg, 25.8 mm Hg) for DBP. However, the SBP and DBP estimated from the test device significantly differed from the ABPM when the arm was raised. The BP readings were higher with the wrist device compared with the ABPM device (mean $\pm$ SD of SBP: 129 ± 14 vs. 108 ± 14 mm Hg, $P < 0.01$; DBP: 83 ± 13 vs. 64 ± 11 mm Hg, $P < 0.01$).

Pulse wave analysis

PWA estimates BP using the morphology of the arterial pulse waveform.²⁴ The shape of the pulse waveform accounts for several physiologic components that affect the accuracy of BP estimation and are not easily accounted for with other approaches, including the amplitude of the reflected waves from bifurcations within the arteries and the contractile force of the heart.²⁵

Hybrid PTT–PWA approaches have been examined. In a study of 1,328 participants, Luo *et al.* employed a machine learning algorithm that predicted SBP, DBP, and pulse pressure from simultaneously measuring facial blood flow in different locations of the face as detected from a smartphone camera, serving as an example of contactless optical imaging for PPG analysis.²⁶ The authors reported Bland–Altman plots comparing their model's estimations with an FDA-approved continuous, beat-to-beat automatic finger cuff device with a measurement bias of 0.39 mm Hg (95% CI -14.21 , 14.99) for SBP and -0.20 mmHg (95% CI -12.20 mm Hg, 11.80 mm Hg) for DBP. These measurement biases are small. While this may suggest that the mean difference across the sample between the cuffless device and the reference device is small, it is important to note that the confidence intervals are wide. This latter finding implies that for an individual, the difference between the test and reference devices may be large, which is problematic for managing patients in clinical practice.

Additional methods

PPG waveforms have been obtained using smartphone cameras. Chandrasekhar *et al.* designed a smartphone-based device that represents an extension of the oscillometric principle of BP measurement.²⁷ The device requires the individual to use his/her finger and apply variable pressure to a PPG sensor located on top of a force transducer. The device application then computes a mean arterial pressure and derives an SBP and DBP estimation based on the changes in amplitude of blood volume oscillation as a function of changes in the amount of pressure applied to the sensor.

LIMITATIONS OF CUFFLESS TECHNOLOGY

The engineering limitations of cuffless technologies are summarized in Table 2. As discussed above, none of the cuffless BP technologies directly measure BP. Each of the variables used to estimate BP and their associations with one another vary between individuals. Yet, each cuffless BP technology makes assumptions about physiologic conditions that affect the BP estimate. For instance, the calibration curve relating PTT to BP is dependent on distance between waveform measurement sites, blood density, average cross-sectional area of arteries between sites, and an individual's association of compliance with BP.²⁸ PPG sensors have not been properly validated across important subject-specific characteristics including skin tone,^{29–31} and many studies have not validated cuffless BP devices among participants with different skin tones. Further, all cuffless devices currently require individual calibration to a cuff-based BP device, which typically uses the oscillometric method. There is limited data available comparing cuffless devices to invasive BP measurement as a reference standard. Calibrating a cuffless device with an oscillometric device, which itself estimates BP using proprietary algorithms, inherently is problematic as the resultant cuffless BP estimate will incorporate the measurement error of the oscillometric device. In addition, most cuff-based devices have been tested to estimate BP in a motionless resting position and not throughout activity or a range of physical positions. Calibration with a cuff-based device at rest may not be appropriate if the cuffless device will be used during activity and different body positions.

Position and movement may also affect cuffless devices accuracy. Several of the aforementioned devices have been tested with the sensor in a fixed body position. However, the BP estimates from these devices are dependent on sensor position. Further, motion artifact during regular activity may degrade the measured signal. If a goal of cuffless BP devices is to provide a means for out-of-office BP monitoring during daily activities, the device needs to be validated in multiple positions and with the same activity level expected outside of the office. While this limitation is not specific to cuffless devices, it must be addressed if cuffless devices are going to provide benefit over the already-available cuff-based devices.

Another limitation of cuffless devices is the drift phenomenon, which is defined as the deviation over time of the device's estimation of BP away from the reference standard used for calibration. Many of the existing validation studies of cuffless devices have tested devices immediately following

Table 2. Engineering limitations of cuffless BP technologies

No direct measurement of blood pressure
Require calibration to a reference standard (e.g. an oscillometric device with BP cuff)
Requires assumption of physiologic conditions (i.e. blood density, cross-sectional area of arteries, vascular stiffness)
PPG sensors are not validated for diverse skin tones
Drift phenomenon
May not detect diurnal BP changes
Accuracy affected by sensor position and movement
Limited study among special populations

calibration; however, these devices drift from the reference standard in days to months, depending on the device.^{32–34} For instance, the BioBeat watch recommends repeating calibration with a cuff-based device every 3 months¹⁶; while the BPro tonometry watch requires calibration before each ABPM session.²³ The frequent need for recalibration can make it challenging for a patient to use cuffless devices at home over an extended period.³⁵

Finally, similar to cuff-based BP devices, there are limited data to determine whether cuffless BP devices are accurate for individuals from special populations, such those that have an upper arm circumference >50–52 cm, coronary artery disease, heart failure, pregnant women (in whom BP physiology may be different), and those with arrhythmias where the beat-to-beat changes in blood volume ejection and electrocardiographic intervals may affect the variables needed for BP estimation.

Due to these concerns about accuracy and usefulness, US and international HTN guidelines have recommended that these devices should not be used in clinical practice.^{1,36} Further studies are required to improve upon the accuracy of these technologies. A recent simulation performed by Mukkamala *et al.* highlights a “paradox” of BP estimation by cuffless devices: BP estimates from a cuffless device may be more accurate when predicted by calibration BP, mean BP of a similar population, and/or demographics (i.e., age and sex) than when based on the cuffless device’s measurement itself (e.g., PTT).³⁷

THE CURRENT STATE OF VALIDATION OF CUFFLESS DEVICES

The 2017 American College of Cardiology (ACC)/American Heart Association (AHA) BP guideline recommends the use of validated BP devices to determine BP for the diagnosis and management of HTN.³⁵ Validated devices have been tested for accuracy against a suitable reference standard, including the auscultatory approach with an upper arm BP cuff and mercury sphygmomanometer or, less commonly, invasive BP measurement. However, invasive BP measurement is not used in outpatient clinical practice, and many prior studies have validated devices to a mercury sphygmomanometer. Currently, many cuffless devices have entered the market and been sold without undergoing independent validation.³⁸ For the few studies in which a cuffless device has been validated, traditional protocol criteria

for cuff-based devices have been used,³⁹ including those from the American National Standards Institute (ANSI)/Association for the Advancement of Medical Instruments (AAMI)/International Organization for Standardization (ISO) validation protocol.⁴⁰

An alternative validation approach would be to compare the associations between cuffless device BP estimations versus traditional office-based BP measurements and cardiovascular disease outcomes. This outcomes-based method would establish the validity of these devices in their role in BP management for cardiovascular risk reduction. However, at present, as there is no outcomes data demonstrating that cuffless devices predict cardiovascular outcome endpoints, further study is needed.

In 2014, the Institute of Electronics and Electrical Engineers (IEEE) published the first validation protocol for cuffless devices.⁴¹ The IEEE protocol includes 2 phases of validation and has requirements for observer training and measurement, participant selection, validation criteria, and data reporting (Supplementary Methods). The IEEE validation protocol differs in some important ways from the traditional protocols to evaluate cuff-based devices (Supplemental Table S1). Of note, this standard recommended that all cuffless devices be calibrated with a mercury sphygmomanometer. Oscillometric devices estimate BP with a proprietary algorithm that accepts a degree of error in comparison to the reference standard, and therefore calibration to these devices would incorporate this error in addition to the error from the cuffless device. Therefore, oscillometric devices are not an appropriate reference standard. In 2019, the IEEE published an addendum that addressed several potential limitations of the 2014 protocol, including requiring performance evaluation in special populations and increasing the sample size required for standard testing (Supplemental Table S2).⁴² One of the most notable changes is the requirement to evaluate the cuffless device’s performance in different positions (i.e., supine, sitting, and standing) and in the presence of motion with varying degrees of noise artifact.

Despite these amendments, limitations remain in the 2019 IEEE standard. The standard does not explicitly require evaluation for drift phenomenon over time. Also, while the amendment does state that devices should be evaluated during all ambulatory conditions or patient activities in which the device could be used and that this testing should be explicitly described in the device instructions, specific dynamic testing protocols are not defined. For example, it is not specified to test device performance in the presence of stressors such as exercise or cold or hot environmental temperatures. Furthermore, the document identifies the issue that there is no currently accepted reference standard for comparison for cuffless devices undergoing dynamic testing. Current oscillometric cuff-based devices are not a suitable reference standard during movement, and even invasive BP monitoring is affected by position.⁴¹ Improved testing for inter- and intraindividual BP variations is necessary for cuffless devices to be considered appropriate for their intended use.³⁷ An AAMI/ISO validation protocol, which is considered the standard for cuff-based devices, has not yet been published for cuffless devices.

U.S. FOOD AND DRUG ADMINISTRATION (FDA) CLEARANCE

There has been confusion regarding the FDA approval process for BP devices including cuffless devices.⁴³ While the FDA provides clearance for a device to be sold on the market, it does not designate the device as accurate.⁴⁴ Therefore, BP devices are not FDA approved but rather FDA cleared. Manufacturers of noninvasive BP devices obtain clearance to market their device through a Premarket Notification application detailed in the 510(k) section of the Federal Food, Drug and Cosmetic Act.⁴⁵ Manufacturers are only required to demonstrate that their device is as safe and effective as similar devices on the market, a concept known as “substantial equivalence.” There is no specific requirement that the device has demonstrated accuracy using a specific validation protocol.⁴⁵

The distinction between FDA clearance and formal device validation is important since cuffless BP devices with FDA clearance can enter the marketplace without being accurate, leading to improper diagnosis and management for HTN.

POTENTIAL BARRIERS TO WIDESPREAD IMPLEMENTATION

There are several potential barriers to widespread clinical implementation of cuffless BP technologies (Table 3).

Uncertainty of how cuffless BP devices should be used

The role of cuffless devices in out-of-office BP monitoring is not defined. It is unclear whether cuffless BP devices should be classified as ABPM devices (i.e., automatic measurements are obtained throughout the day and night at set intervals during normal activity including sleep) or HBPM devices (i.e., self-triggered measurements are obtained by a person during the morning and evening in a standardized manner). These devices could also comprise a third category since some cuffless BP devices continuously determine beat-to-beat BP, which is different than most of the current ABPM and HBPM devices. The 2017 ACC/AHA BP guideline published recommendations on what BP thresholds should be used to define high out-of-office BP. These recommendations were based on studies that included traditional ABPM and HBPM devices with an upper arm cuff. It remains unknown if the same BP thresholds be used for cuffless BP devices.

Furthermore, acknowledgement of the intended use and limitations of cuffless devices must be made clear to patients and clinicians to prevent harm. For example, if cuffless BP devices are used outside of the setting for which they were validated, patients and providers using cuffless devices to guide clinical management may misdiagnose mismanage HTN. Overtreatment of a falsely elevated BP may lead to adverse events associated with antihypertensive medication, and undertreatment of a falsely low BP may leave a person at increased cardiovascular risk. This raises the question of whether these devices should be identified as BP estimation devices, given that they do not directly measure BP.

Data storage and use

There may be substantial volume of BP data that is captured by cuffless BP devices, especially if used frequently over time and in a continuous manner. In this scenario, it would be advantageous to have a safe, secure, and cloud-based data storage solution since it may be difficult for patients to store and manage these data locally at home and transmit them to the clinician. Providers will also need a feasible approach to interpret and act on data streams from cuffless BP devices. Both providers and patients need training and education in how to interpret BP trends over time and management strategies for individual BP readings that may be outliers.

Reliability and the need for device calibration

There is limited data regarding the reliability of cuffless BP devices over time. A cuffless BP device that does not manifest a drift phenomenon in calibration must still provide reliable estimates to be useful in clinical practice. Additionally, cuffless BP devices must be designed to have longer periods of use without the need for recalibration to offer practical advantages over current out-of-office BP monitoring approaches.

Reimbursement

Currently, HBPM devices must be purchased out-of-pocket. If the same is true for cuffless BP devices as they emerge on the market, it may exacerbate the widening the gap between people who have access to novel health care technologies and can benefit from these approaches, and those who do not have access.^{46–50}

In conclusion, BP measurement with a BP cuff has been the primacy of HTN diagnosis and management. There are exciting and emerging technologies to estimate BP without a BP cuff, which have great potential for advancing the field of out-of-office BP monitoring and building on the tools available to primary care providers in the management of HTN. However, despite rapid progress and device development in cuffless BP technologies, several limitations and barriers must first be addressed before these technologies can be implemented into clinical practice in a meaningful

Table 3. Potential barriers to widespread use of cuffless BP technologies

Lack of scientific transparency on how BP is estimated
Unclear classification as an ABPM or HBPM device
The BP thresholds for the diagnosis of HTN and BP control using cuffless devices are not known
Large volume of accumulated data
Need for patient and clinician training in interpretation and clinical management of data
Lack of evidence demonstrating patient comfort and ease of use
Unclear target population
Unclear reliability and requirements for device calibration
Issues of reimbursement

way. Without careful attention to these issues, cuffless BP devices may not provide any substantial advantage to patients and clinicians for the diagnosis and management of HTN. Improper use of cuffless BP technologies could lead to misdiagnosis, over- or undertreatment of HTN, and the widening of ongoing disparities in BP control. Before cuffless devices have a role in clinical care, more research, including testing implementation approaches, needs to be conducted.

SUPPLEMENTARY MATERIAL

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